

	Visual	Mean	Color	Mat.	Shape	Size
IEP	Conv.	90.6	91.0	90.0	89.9	90.6
MAC	Attn.	95.9	98.0	91.4	94.4	94.2
TbD (hres.)	Attn.	96.5	96.6	92.2	95.4	92.6
NS-CL	Obj.	98.7	99.0	98.7	98.1	99.1

Table 2: We also evaluate the learned visual concepts using a diagnostic question set containing simple questions such as “How many red objects are there?”. NS-CL outperforms both convolutional and attentional baselines. The suggested object-based visual representation and symbolic reasoning approach perceives better interpretation of visual concepts.

Model	Visual	Accuracy (100% Data)	Accuracy (10% Data)
TbD	Attn.	99.1	54.2
TbD-Object	Obj.	84.1	52.6
TbD-Mask	Attn.	99.0	55.0
MAC	Attn.	98.9	67.3
MAC-Object	Obj.	79.5	51.2
MAC-Mask	Attn.	98.7	68.4
NS-CL	Obj.	99.2	98.9

Table 3: We compare different variants of baselines for a systematic study on visual features and data efficiency. Using only 10% of the training images, our model is able to achieve a comparable results with the baselines trained on the full dataset. See the text for details.

Table 4 summarizes results on the CLEVR validation split. Our model achieves the state-of-the-art performance among all baselines using zero program annotations, including MAC (Hudson & Manning, 2018) and FiLM (Perez et al., 2018). Our model achieves comparable performance with the strong baseline TbD-Nets (Mascharka et al., 2018), whose semantic parser is trained using 700K programs in CLEVR (ours need 0). The recent NS-VQA model from Yi et al. (2018) achieves better performance on CLEVR; however, their system requires annotated visual attributes and program traces during training, while our NS-CL needs no extra labels.

Here, the visual perception module is pre-trained on ImageNet (Deng et al., 2009). Without pre-training, the concept learning accuracies drop by 0.2% on average and the QA accuracy drops by 0.5%. Meanwhile, NS-CL recovers the underlying programs of questions accurately ($> 99.9\%$ accuracy). NS-CL can also detect ambiguous or invalid programs and indicate exceptions. Please see Appendix F for more details. NS-CL can also be applied to other visual reasoning testbeds. Please refer to Appendix G.1 for our results on the Minecraft dataset (Yi et al., 2018).

For a systematic study on visual features and data efficiency, we implement two variants of the baseline models: TbD-Object and MAC-Object. Inspired by (Anderson et al., 2018), instead of the input image, TbD-Object and MAC-Object take a stack of object features as input. TbD-Mask and MAC-Mask integrate the masks of objects by using them to guide the attention over the images.

Table 3 summarizes the results. Our model outperforms all baselines on data efficiency. This comes from the full disentanglement of visual concept learning and symbolic reasoning: how to execute program instructions based on the learned concepts is programmed. TbD-Object and MAC-Object demonstrate inferior results in our experiments. We attribute this to the design of model architectures and have a detailed analysis in Appendix F.3. Although TbD-Mask and MAC-Mask do not perform better than the originals, we find that using masks to guide attentions speeds up the training.

Besides achieving a competitive performance on the visual reasoning testbeds, by leveraging both object-based representation and symbolic reasoning, our model learns fully interpretable visual concepts: see Appendix H for qualitative results on various datasets.

4.3 GENERALIZATION TO NEW ATTRIBUTES AND COMPOSITIONS

Generalizing to new visual compositions. The CLEVR-CoGenT dataset is designed to evaluate models’ ability to generalize to new visual compositions. It has two splits: Split A only contains gray, blue, brown and yellow cubes, but red, green, purple, and cyan cylinders; split B imposes the opposite color constraints on cubes and cylinders. If we directly learn visual concepts on split A, it overfits to classify shapes based on the color, leading to a poor generalization to split B.

Our solution is based on the idea of seeing attributes as operators. Specifically, we jointly train the concept embeddings (e.g., Red, Cube, etc.) as well as the semantic parser on split A, keeping pre-trained, frozen attribute operators. As we learn distinct representation spaces for different attributes, our model achieves an accuracy of 98.8% on split A and 98.9% on split B.

Model	Prog. Anno.	Overall	Count	Cmp. Num.	Exist	Query Attr.	Cmp. Attr.
Human	N/A	92.6	86.7	86.4	96.6	95.0	96.0
NMN	700K	72.1	52.5	72.7	79.3	79.0	78.0
N2NMN	700K	88.8	68.5	84.9	85.7	90.0	88.8
IEP	700K	96.9	92.7	98.7	97.1	98.1	98.9
DDRprog	700K	98.3	96.5	98.4	98.8	99.1	99.0
TbD	700K	99.1	97.6	99.4	99.2	99.5	99.6
RN	0	95.5	90.1	93.6	97.8	97.1	97.9
FiLM	0	97.6	94.5	93.8	99.2	99.2	99.0
MAC	0	98.9	97.2	99.4	99.5	99.3	99.5
NS-CL	0	98.9	98.2	99.0	98.8	99.3	99.1

Table 4: Our model outperforms all baselines using no program annotations. It achieves comparable results with models trained by full program annotations such as TbD.

Model	Test			
	Split A	Split B	Split C	Split D
MAC	97.3	N/A	92.9	N/A
IEP	96.1	92.1	91.5	90.9
TbD	98.8	94.5	94.3	91.9
NS-CL	98.9	98.9	98.7	98.8

Figure 5: We test the combinatorial generalization w.r.t. the number of objects in scenes and the complexity of questions (i.e. the depth of the program trees). We make four splits of the data containing various complexities of scenes and questions. Our object-based visual representation and explicit program semantics enjoys the best (and almost-perfect) combinatorial generalization compared with strong baselines.

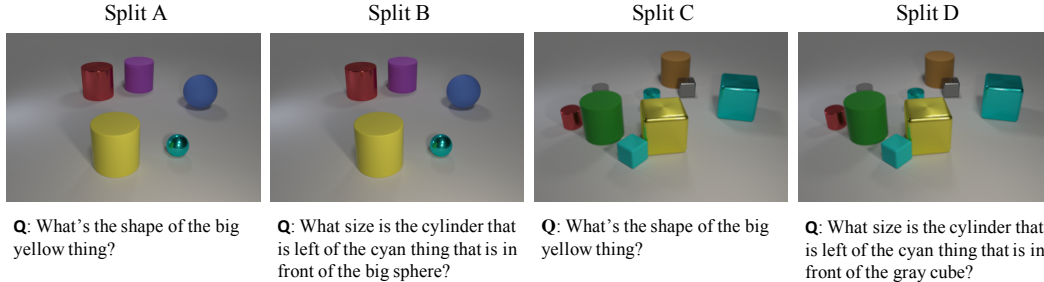


Figure 6: Samples collected from four splits in Section 4.3 for illustration. Models are trained on split A but evaluated on all splits for testing the combinatorial generalization.

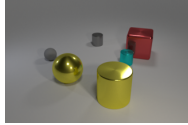
Generalizing to new visual concepts. We expect the process of concept learning can take place in an incremental manner: having learned 7 different colors, humans can learn the 8-th color incrementally and efficiently. To this end, we build a synthetic split of the CLEVR dataset to replicate the setting of incremental concept learning. Split A contains only images without any purple objects, while split B contains images with at least one purple object. We train all the models on split A first, and finetune them on 100 images from split B. We report the final QA performance on split B’s validation set. All models use a pre-trained semantic parser on the full CLEVR dataset.

Our model performs a 93.9% accuracy on the QA test in Split B, outperforming the convolutional baseline IEP (Johnson et al., 2017b) and the attentional baseline TbD (Mascharka et al., 2018) by 4.6% and 6.1% respectively. The acquisition of `color` operator brings more efficient learning of new visual concepts.

4.4 COMBINATORIAL GENERALIZATION TO NEW SCENES AND QUESTIONS

Having learned visual concepts on small-scale scenes (containing only few objects) and simple questions (only single-hop questions), we humans can easily generalize the knowledge to larger-scale scenes and to answer complex questions. To evaluate this, we split the CLEVR dataset into four parts: **Split A** contains only scenes with less than 6 objects, and questions whose latent programs having a depth less than 5; **Split B** contains scenes with less than 6 objects, but arbitrary questions; **Split C** contains arbitrary scenes, but restricts the program depth being less than 5; **Split D** contains arbitrary scenes and questions. Figure 6 shows some illustrative samples.

As VQA baselines are unable to count a set of objects of arbitrary size, for a fair comparison, all programs containing the “count” operation over > 6 objects are removed from the set. For



Caption: There is a big yellow cylinder in front of a gray object.

(a) An illustrative pair of image and caption in our synthetic dataset.

Model	Retrieval Accuracy
IEP	95.5
TbD	97.0
NS-CL	96.9

(b) Image-caption retrieval accuracy on a subset of data. Our model archives comparable results with VQA baselines.

Model	Retrieval Accuracy
CNN-LSTM	68.9
NS-CL	97.0

(c) Image-caption retrieval accuracy on the full dataset. Our model outperforms baselines and requires no extra training or fine-tuning of the visual perception module.

Table 5: We introduce a new simple DSL for image-caption retrieval to evaluate how well the learned visual concepts transfer. Due to the difference between VQA and caption retrieval, VQA baselines are only able to infer the result on a partial set of data. The learned object-based visual concepts can be directly transferred into the new domain for free.

methods using explicit program semantics, the semantic parser is pre-trained on the full dataset and fixed. Methods with implicit program semantics (Hudson & Manning, 2018) learn an entangled representation for perception and reasoning, and cannot trivially generalize to more complex programs. We only use the training data from the Split A and then quantify the generalization ability on other three splits. Shown in Table 5, our NS-CL leads to almost-perfect generalization to larger scenes and more complex questions, outperforming all baselines by at least 4% in QA accuracy.

4.5 EXTENDING TO OTHER PROGRAM DOMAIN

The learned visual concepts can also be used in other domains such as image retrieval. With the visual scenes fixed, the learned visual concepts can be directly transferred into the new domain. We only need to learn the semantic parsing of natural language into the new DSL.

We build a synthetic dataset for image retrieval and adopt a DSL from scene graph-based image retrieval (Johnson et al., 2015). The dataset contains only simple captions: “There is an <object A> <relation> <object B>.” (e.g., There is a box right of a cylinder). The semantic parser learns to extract corresponding visual concepts (e.g., `box`, `right`, and `cylinder`) from the sentence. The program can then be executed on the visual representation to determine if the visual scene contains such relational triples.

For simplicity, we treat retrieval as classifying whether a relational triple exists in the image. This functionality cannot be directly implemented on the CLEVR VQA program domain, because questions such as “Is there a box right of a cylinder” can be ambiguous if there exist multiple cylinders in the scene. Due to the entanglement of the visual representation with the specific DSL, baselines trained on CLEVR QA can not be directly applied to this task. For a fair comparison with them, we show the result in Table 5b on a subset of the generated image-caption pairs where the underlying programs have no ambiguity regarding the reference of object B. A separate semantic parser is trained for the VQA baselines, which translates captions into a CLEVR QA-compatible program (e.g., `Exist(Filter(Box, Relate(Right, Filter(Cylinder))))`).

Table 5c compares our NS-CL against typical image-text retrieval baselines on the full image-caption dataset. Without any annotations of the sentence semantics, our model learns to parse the captions into the programs in the new DSL. It outperforms the CNN-LSTM baseline by 30%.

4.6 EXTENDING TO NATURAL IMAGES AND LANGUAGE

We further conduct experiments on MS-COCO (Lin et al., 2014) images. Results are presented on the VQS dataset (Gan et al., 2017). VQS contains a subset of images and questions from the original VQA 1.0 dataset (Antol et al., 2015). All questions in the VQS dataset can be visually grounded: each question is associated with multiple image regions, annotated by humans as essential for answering the question. Figure 7 illustrates an execution trace of NS-CL on VQS.

We use a syntactic dependency parser to extract programs and concepts from language (Andreas et al., 2016; Schuster et al., 2015). The object proposals and features are extracted from models pre-trained on the MS-COCO dataset and the ImageNet dataset, respectively. Illustrated in Figure 7, our model